SS-P-196E
July 20, 1978
SUPERSEDING
Int. Fed. Spec. SS-P-00196D(GSA-FSS)
April 26, 1974, and
Fed. Spec. SS-P-196C
June 20, 1960

FEDERAL SPECIFICATION

PENCIL: SPIRAL-PAPER-FORM; WAX, GLASS AND CHINA MARKING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers paper-wrapped wax marking core pencils.

1.2 Classification.

1.2.1 Types. The pencils shall be of the following types, as specified (see 6.2).

Type I - Thick marking core.
Type II - Thin marking core.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.

(Activities outside the Federal Government may obtain copies of Federal Specifications Standards and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 7510

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National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Reclaimed material. The paper wrap shall contain not less than 20 percent of recovered materials. Recovered materials are defined as material which has been collected or recovered from solid waste. Solid waste is defined as any garbage, refuse, sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility and other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations and from community activities, but does not include solid or dissolved material in domestic sewage, or solid or dissolved materials in irrigation return flows or industrial discharges.

3.1.2 Marking core. The marking core shall be a mixture of extender, wax, and pigments compounded to meet the requirements of this specification.

3.1.3 Wrapping. The wrapping shall be paper of the type normally used in the commercial production of the pencils covered by this specification and which shall insure conformance to the requirements specified herein.

3.2 Construction.

3.2.1 Wrapping. The marking core shall be spirally wrapped with a minimum of seventeen layers of paper (see 3.1.3). The wrapping shall be scored in accordance with commercial practice to permit removal of a strip of paper without tearing or delamination to expose the marking core as it is used.(see 4.4.4). Each of the paper strips shall be provided with a suitable method for starting the removal process of the paper wrap.

3.2.2 Marking core. The marking core shall be in one piece and shall be securely held throughout the entire length of the paper wrap to prevent slippage of the core during normal usage (see 4.4.4). The marking core shall be of the color specified (see 6.2).

3.3 Dimensions.

3.3.1 Pencils. Pencils shall be not less than 6-1/4 inches long with an outside diameter of 0.335 $\pm$ 0.040 inch.

3.3.2 Marking core. The diameter of the marking core shall be 0.185 to 0.200 inch for type I and 0.150 to 0.165 for type II pencils.

3.4 Performance.

3.4.1 Marking core breaking strength. The marking core shall have a breaking strength of not less than 2100 grams for type I pencils and 1450 grams for type II pencils when tested in accordance with 4.4.1.

3.4.2 Legibility and stability of marking core. Core marking shall be clearly legible, with no visible color change, and shall not run when tested in accordance with 4.4.2.

3.4.3 Marking characteristics. The marking core shall leave a continuous, clearly visible mark of the color specified, free from chips and crumbs, when tested as specified in 4.4.3.

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3.5 Finish. The outer layer of the wrapping shall be finished to represent the color of the marking core in accordance with normal commercial practice.

3.6 Identification marking. Each pencil shall be clearly marked with the manufacturer's name or trademark and the marking core color. Additional identifying marking consistent with commercial practice may also be imprinted.

3.7 Workmanship. The pencils shall be free from any defects which may affect appearance or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.2.1 Sampling for visual and dimensional examination. The sample unit shall be one pencil. The Inspection Level shall be S-2, and the Acceptable Quality Level (AQL) shall be 4.0 percent defective.

4.2.2 Sampling for examination of preparation for delivery. The sample unit shall be one shipping container. The Inspection Level shall be S-2, and the AQL shall be 4.0 percent defective.

4.2.3 Sampling for test. The sample unit shall be two pencils. The Inspection Level shall be S-2, and the AQL shall be 2.5 percent defective.

4.3 Examination.

4.3.1 Visual and dimensional examination. Samples selected in accordance with 4.2.1 shall be examined to determine that the pencils conform to Section 3 of this specification.

4.3.2 Examination of preparation for delivery. Samples selected in accordance with 4.2.2 shall be examined to determine that packaging, packing and marking requirements of Section 5 of this specification are complied with.

4.4 Tests. Samples taken in accordance with 4.2.3 shall be subjected to the following tests.

4.4.1 Marking core breaking strength. Test shall be conducted in an atmosphere maintained at a temperature of 23° ± 2°C. (73.5° ± 3.5°F.), relative humidity 50 ± 2 percent. Pencils shall be conditioned in these atmospheric conditions for a period of 12 hours before the test.

Test shall be conducted using a platform scale and any suitable holding device which will maintain the pencil at a 45° angle to the scale platform and which will permit uniform downward pressure of the holder and pencil. The scale platform shall be smooth.

Pencils will be prepared so that a square-cut, unsharpened tip of the marking core extends 11/64 ± 1/64 inch measured from the lowest point of the paper wrap.

The pencil shall be secured in the holder with the pencil extended 1-1/4 to 1-1/2 inches beneath the underside of the holder, measured along the pencil to the prepared end, and shall touch the scale platform. Downward pressure shall be applied uniformly. Readings shall be taken at the instant of breaking. See paragraph 3.4.1.

4.4.2 Legibility and stability. Inscribe six 1/2 inch high letters and numbers on the side of a glass beaker with a sample pencil. Place the beaker in an oven at a temperature of 300° ± 5°F. for a period of 1 hour. The beaker shall be removed and allowed to cool to room temperature. Markings shall be examined for conformance with 3.4.2.

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4.4.3 Marking characteristics. The same pencils used in the preceding tests shall be tested for writing on the following glossy surface materials normally found in a laboratory: glass, aluminum, including aluminum foil, cellophane and cellulose acetate tape, and glazed white porcelain. Straight lines and zig-zag lines consisting of short straight lines forming sharp angles shall be drawn on each of the specified materials. The markings shall then be examined for compliance with 3.4.3.

4.4.4 Wrapping and marking core. After the pencil has been used as specified in 4.4.2 and 4.4.3, examine the pencil to determine whether usage resulted in core slippage (see 3.2.2). Completely remove the wrapper a strip at a time and examine for proper removal (see 3.2.1) and examine to determine whether the core is in one piece (see 3.2.2).

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Twelve pencils of one color shall be packaged in a paperboard open-end sleeve, clear plastic tray or other suitable container.

5.1.1.2 Intermediate packaging. Six unit packages of pencils shall be intermediate packaged in a close-fitting paperboard box. The box shall be securely closed to prevent accidental opening.

5.1.2 Commercial. The pencils shall be packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling and storage.

5.2 Packing. Packing shall be Level A or Commercial, as specified (see 6.2).

5.2.1 Level A. Twenty intermediate packaged pencils (120 unit packages), packaged as specified in 5.1, shall be packed in close-fitting fiberboard boxes conforming to PPP-B-636, class V3c. The boxes shall be closed, waterproofed, and reinforced in accordance with the appendix to PPP-B-636. Alternatively, wirebound, cleated plywood or nailed wood boxes shall be acceptable shipping containers when lined with a waterproof barrier material. The barrier material shall be sealed at the edges with waterproof tape or adhesive.

5.2.2 Commercial. The pencils, packaged as specified in 5.1, shall be packed in fiberboard boxes that will assure acceptance by common carrier, and provide product protection against loss and damage during multiple shipments, handling and storage. The shipping container shall be in compliance with the National Motor Freight Classification or Uniform Freight Classification.

6. NOTES

6.1 Intended use. Pencils covered by this specification are intended for marking on a variety of glossy surfaces and may be used for marking glass and apparatus subject to oven or Bunsen burner heat up to 300°F.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type and color required (see 1.2.1 and 3.2.2).
(c) Inspection and testing responsibility, if other than specified in 4.1.
(d) Level of packaging and packing required (see 5.1 and 5.2).

Preparing Activity	Military Interest
GSA-FSS	DOD HAS WAIVED COORDINATION

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 50 cents each.

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4.2.3 Examination of preparation for delivery. Samples taken in accordance with 4.2.3